Chapter 4
Limits and Continuity

Note 3

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4.3 : Continuity

Recall that, a function is called continuous at a if $a \in \text{Dom } f$ and $f(x) \to f(a)$ as $x \to a$. In particular, it is assumed that f is defined on BOTH sides of a. Here, we introduce a more general concept of continuity which includes functions, such as $\sqrt{a}$ at $a = 0$, which are defined on only one side of some point in their domain.

Definition 4.3.1 : Let E be a non-empty subset of $\mathbb{R}$ and $f : E \to \mathbb{R}$.

(i). f is said to be *continuous at a point* $a \in E$ if and only if given $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|x - a| < \delta \quad \text{and} \quad x \in E \quad \text{imply} \quad |f(x) - f(a)| < \varepsilon \tag{5}$$

(ii). f is said to be *continuous on* E (notation: $f : E \to \mathbb{R}$ is continuous) if and only if f is continuous at every $x \in E$.

Remark : Let I be an open interval which contains a point a and $f : I \to \mathbb{R}$. Then f is continuous at $a \in I$ if and only if $f(a) = \lim_{x \to a} f(x)$

Proof of the Remark:

- Suppose that $I = (c, d)$ and set $\delta_0 := \min\{|c - a|, \ |d - a|\}$. If $\delta < \delta_0$, then $|x - a| < \delta$ implies $x \in I$.
- Therefore, condition (5) is identical to (1) when $f(a) = L$, $E = I$ and $\delta < \delta_0$.
- It follows that f is continuous at $a \in I$ if and only if $f(x) \to f(a)$ as $x \to a$.

□

By repeating the proof of **Theorem 4.1.1**, we can establish a **Sequential Characterization of Continuity** which is valid on any non-empty set.

Theorem 4.3.1 : Suppose that E is a non-empty subset of $\mathbb{R}$, that $a \in E$ and that $f : E \to \mathbb{R}$. Then the following statements are equivalent:

(i). f is continuous at $a \in E$.

(ii). If $\{x_n\}$ converges to a and $x_n \in E$, then $f(x_n) \to f(a)$ as $n \to \infty$.

By combining **Theorem 4.3.1** with **Theorem 3.2.3**, we obtain the following result.

Theorem 4.3.2 : Let E be a non-empty subset of $\mathbb{R}$ and $f, g : E \to \mathbb{R}$. If f, g are continuous at a point $a \in E$ (respectively, continuous on the set E), then so are $f + g$, fg, and αf (for any $\alpha \in \mathbb{R}$). Moreover, f/g is continuous at $a \in E$ when $g(a) \neq 0$ (respectively, on E when $g(x) \neq 0$ for all $x \in E$).

Remark :

- If f, g are continuous at a point $a \in E$ or on a set E, then so are $|f|, f^+, f^-, f \vee g$ and $f \wedge g$.
- Every polynomial is continuous on $\mathbb{R}$.
- Many complicated functions can be broken into simpler pieces, using sums, products, quotients, and the following operation.

Definition 4.3.2 : Suppose that A and B are subsets of $\mathbb{R}$, that $f : A \to \mathbb{R}$ and $g : B \to \mathbb{R}$. If $f(A) \subseteq B$, then the composition of g with f is the function $g \circ f : A \to \mathbb{R}$ defined by

$$(g \circ f)(x) \coloneqq g(f(x)), \quad x \in A.$$

The following result contains information about when a limit sign and something else (in this case, the evaluation of a function) can be interchanged. Throughout the Analysis course sequence, you will encounter this theme many times, identifying conditions under which you can interchange any two of the following: limits, integrals, derivatives, infinite summations, and computation of a function.

Theorem 4.3.3 : Suppose that A and B are subsets of $\mathbb{R}$, that $f : A \to \mathbb{R}$ and $g : B \to \mathbb{R}$, and that $f(x) \in B$ for every $x \in A$.

(i). If $A := I\backslash\{a\}$, where I is a non-degenerate interval which either contains a or has a as one of its endpoints, if

$$L := \lim_{\substack{x \to a \\ x \in I}} f(x)$$

exists and belongs to B, and if g is continuous at $L \in B$, then

$$\lim_{\substack{x \to a \\ x \in I}} (g \circ f)(x) = g\left(\lim_{\substack{x \to a \\ x \in I}} f(x)\right).$$

(ii). If f is continuous at $a \in A$ and g is continuous at $f(a) \in B$, then $g \circ f$ is continuous at $a \in A$.

Proof of Theorem 4.3.3.

- Suppose that $\{x_n\} \in I\backslash\{a\}$ and that $x_n \to a$ as $n \to \infty$.
- Since $f(A) \subseteq B,\ f(x_n) \in B$.
- Also, by the **Sequential Characterization of Limits (Theorem 4.1.1)**, $f(x_n) \to L$ as $n \to \infty$.
- Since g is continuous at $L \in B$, it follows from **Theorem 4.3.1** that $g \circ f(x_n) \coloneqq g(f(x_n)) \to g(L)$ as $n \to \infty$.
- Hence, by **Theorem 4.2.2**, $g \circ f(x) \to g(L)$ as $x \to a$ in I. This proves **(i)**.
- A similar proof establishes part **(ii)**.

□

For many applications, it is important to be able to find the maximum or minimum of a given function. As a first step in this direction, we introduce the following concept.

Definition 4.3.3 : Let E be a non-empty subset of $\mathbb{R}$. A function $f : E \to \mathbb{R}$ is said to be *bounded* on E if and only if there is an $M > 0$ such that $|f(x)| \leq M$ for all $x \in E$.

Notice that whether a function f is bounded or not on a set E depends on E as well as on f. For example,

(i). $f(x) = \dfrac{1}{x}$ is dominated by 1 on $[1, \infty)$ but unbounded on $(0, 2)$.

(ii). $f(x) = x^2$ is dominated by 4 on $(-2, 2)$ but unbounded on $[0, \infty)$.

The following result, which will be used often, shows that a continuous function on a closed, bounded interval is always bounded.

Theorem 4.3.4 : Extreme Value Theorem (EVT)

If I is a closed, bounded interval and $f : I \to \mathbb{R}$ is continuous on I, then f is bounded on I. Moreover, if

$$M = \sup_{x \in I} f(x) \quad \text{and} \quad m = \inf_{x \in I} f(x),$$

then there exists points $x_m,\ x_M \in I$ such that

$$f(x_M) = M \quad \text{and} \quad f(x_m) = m\ . \tag{6}$$

Proof of Theorem 4.3.4.

- Suppose first that f is not bounded on I. Then there exist $x_n \in I$ such that

$$|f(x_n)| > n, \quad n \in \mathbb{Z}^+\ . \tag{7}$$

- Since I is bounded, we know (by the **Bolzano-Weierstrass Theorem**) that $\{x_n\}$ has a convergent subsequence, say $x_{n_k} \to a$ as $k \to \infty$.

- Since I is closed, we also know (by the **Comparison Theorem**) that $a \in I$. In particular, $f(a) \in \mathbb{R}$.

- On the other hand, substituting n_k for n in (7) and taking the limit of this inequality as $k \to \infty$, we have $|f(a)| = \infty$, a contradiction. Hence f is bounded on I. (We have proved that both M and m are finite real numbers.)

- To show that there is an $x_M \in I$ such that $f(x_M) = M$, suppose to the contrary that $f(x) < M$ for all $x \in I$.
- Then the function $g(x) = \dfrac{1}{M - f(x)}$ is continuous, hence bounded on I. In particular, there is a $C > 0$ such that $|g(x)| = g(x) \leq C$.
- It follows that

$$f(x) \leq M - \frac{1}{C} \quad \text{for all} \quad x \in I \ . \tag{8}$$

- Taking the supremum of (8) over all $x \in I$, we obtain $M \leq M - \dfrac{1}{C} < M$, a contradiction.
- Hence, there is an $x_M \in I$ such that $f(x_M) = M$.
- A similar argument proves that there is an $x_m \in I$ such that $f(x_m) = m$.

□

Remark :

- Another way to interpret (6) is by saying that supremum and infimum of f are *attained* on I.
- The value M (respectively, m) is called the *maximum* (respectively, *minimum*) of f on I.
- Neither of the hypotheses on the interval I in **Theorem 4.3.4** can be relaxed. i.e., The EVT is false if either **closed** or **bounded** is dropped from the hypotheses.
  - The interval $(0, 1)$ is bounded but not closed, and the function $f(x) = \dfrac{1}{x}$ is continuous and unbounded on $(0, 1)$.
  - The interval $[0, \infty)$ is closed but not bounded, and the function $f(x) = x$ is continuous and unbounded on $[0, \infty)$.

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